Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

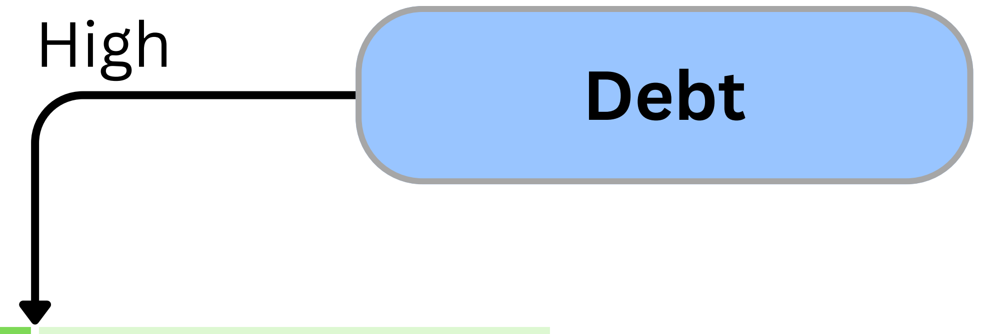
Finding the best split for the root node:

- We need to pick **1 feature** among **Debt, Income, Age** to be at the top of the tree.
- This can be done by calculating the **Gini impurity** for **each feature**.

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 1: Calculate the **Gini impurity** for **feature Debt**.

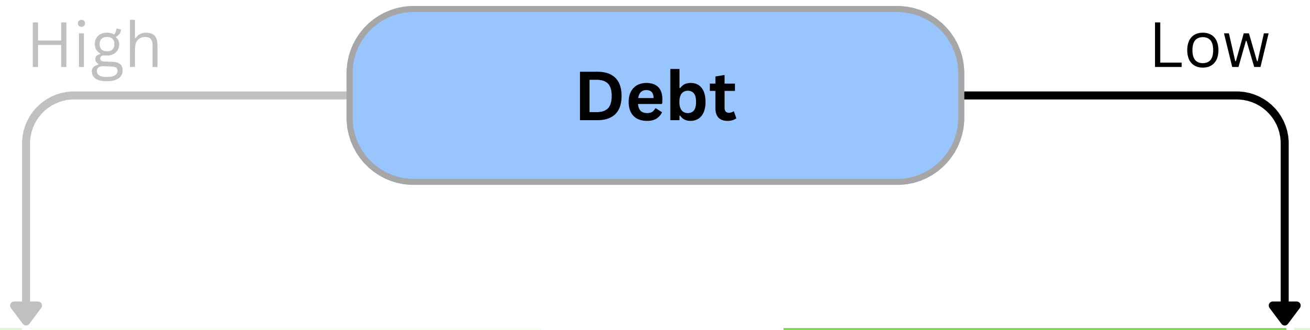


Homeowners	1
Not homeowners	3
Gini impurity	$= 1 - (1/4)^2 - (3/4)^2$ $=$ **0.375**

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 1: Calculate the **Gini impurity** for **feature Debt**.



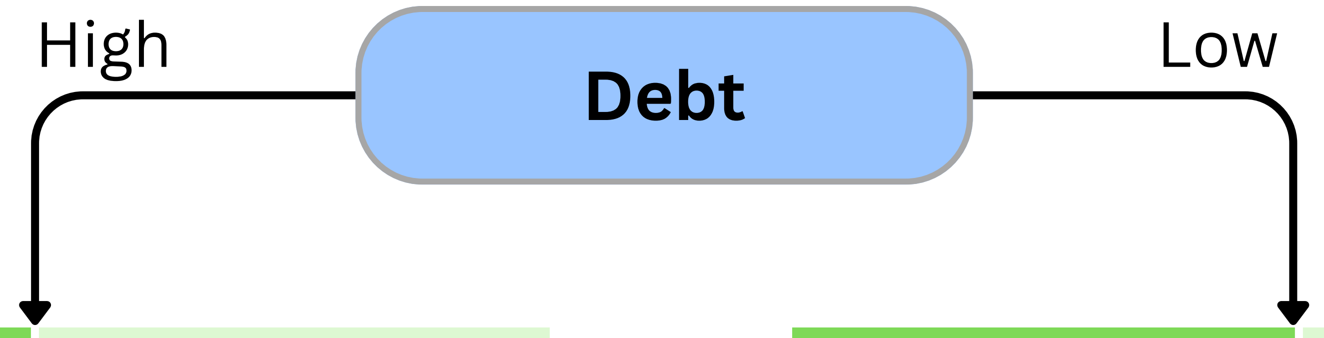
Homeowners	1
Not homeowners	3
Gini impurity	$= 1 - (1/4)^2 - (3/4)^2$ $=$ **0.375**

Homeowners	2
Not homeowners	1
Gini impurity	$= 1 - (2/3)^2 - (1/3)^2$ $=$ **0.444**

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 1: Calculate the **Gini impurity** for **feature Debt**.



Homeowners	1
Not homeowners	3
Gini impurity	$= 1 - (1/4)^2 - (3/4)^2$ $=$ **0.375**

Homeowners	2
Not homeowners	1
Gini impurity	$= 1 - (2/3)^2 - (1/3)^2$ $=$ **0.444**

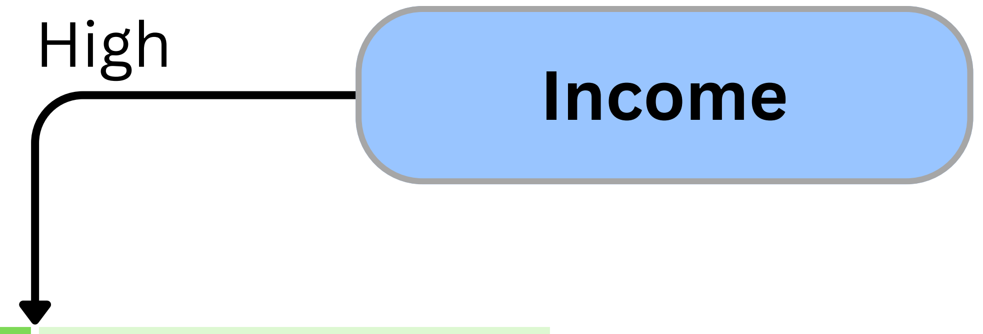
Weighted Gini impurity for feature **Debt**

$= (4/7) \times 0.375 + (3/7) \times 0.444 =$ **0.405**

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 2: Calculate the **Gini impurity** for **feature Income**.

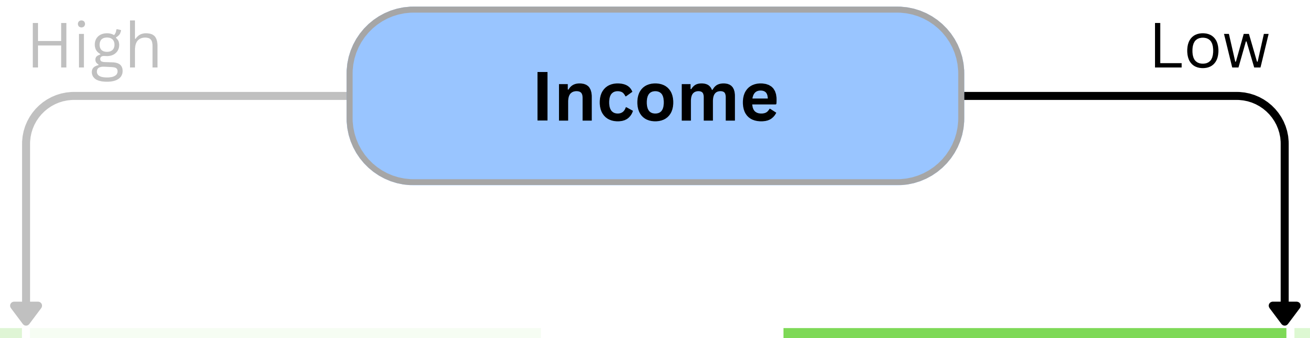


Homeowners	3
Not homeowners	1
Gini impurity	$= 1 - (3/4)^2 - (1/4)^2$ $= \mathbf{0.375}$

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

Step 2: Calculate the **Gini impurity** for **feature Income**.



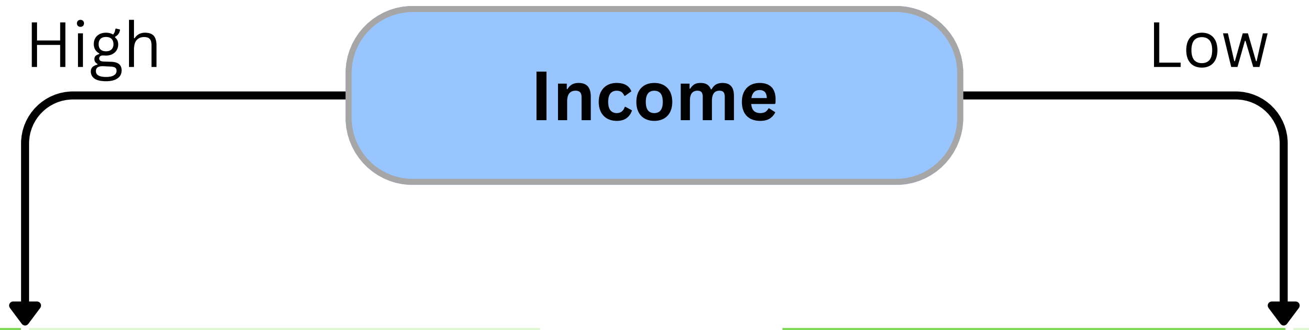
Homeowners	3
Not homeowners	1
Gini impurity	$= 1 - (3/4)^2 - (1/4)^2$ $= \mathbf{0.375}$

Homeowners	0
Not homeowners	3
Gini impurity	$= 1 - (0/3)^2 - (3/3)^2$ $= \mathbf{0.0}$

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 2: Calculate the **Gini impurity** for **feature Income**.



Homeowners	3
Not homeowners	1
Gini impurity	$= 1 - (3/4)^2 - (1/4)^2$ $=$ **0.375**

Homeowners	0
Not homeowners	3
Gini impurity	$= 1 - (0/3)^2 - (3/3)^2$ $=$ **0.0**

Weighted Gini impurity for feature **Income**

$= (4/7) \times 0.375 + (3/7) \times 0.0 =$ **0.214**

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 3: Calculate the **Gini impurity** for **feature Age**.

- First, we **sort** the data by age **in ascending order** (in this case, it’s already sorted)
- Next, we calculate the average age of each pair of adjacent people.
- Finally, we calculate the **Gini impurity for each Age threshold**.

Debt	Income	Age	Home owner
High	High	7	No
		(7 + 12) / 2 = 9.5	
High	Low	12	No
		15	
Low	High	18	Yes
		26.5	
Low	High	35	Yes
		36.5	
High	High	38	Yes
		44	
High	Low	50	No
		66.5	
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

(7 + 12) / 2 = 9.5 | 15 | 26.5 | 36.5 | 44 | 66.5

Step 3: Calculate the **Gini impurity** for **feature Age.**

An example below shows how to calculate **Gini impurity** for **threshold Age < 44**

Age < 44

True

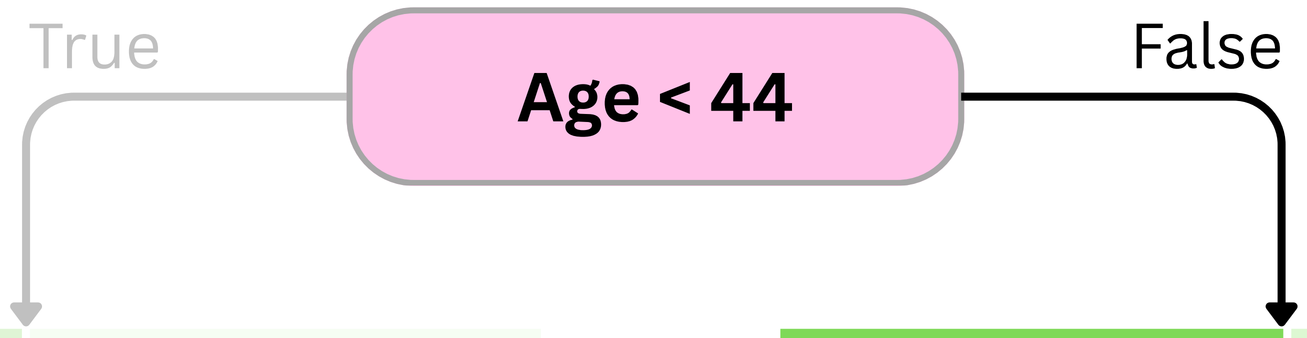
Homeowners	3
Not homeowners	2
Gini impurity	$= 1 - (3/5)^2 - (2/5)^2$ $=$ **0.48**

Debt	Income	Age	Home owner
High	High	7	No
		(7 + 12) / 2 = 9.5	
High	Low	12	No
		15	
Low	High	18	Yes
		26.5	
Low	High	35	Yes
		36.5	
High	High	38	Yes
		44	
High	Low	50	No
		66.5	
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 3: Calculate the **Gini impurity** for **feature Age.**

An example below shows how to calculate **Gini impurity** for **threshold Age < 44**



Homeowners	3
Not homeowners	2
Gini impurity	$= 1 - (3/5)^2 - (2/5)^2$ = **0.48**

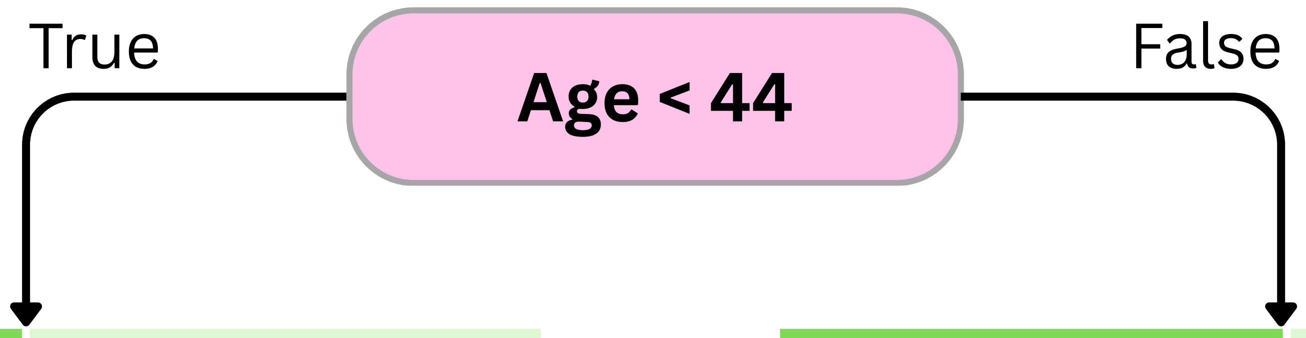
Homeowners	0
Not homeowners	2
Gini impurity	$= 1 - (0/2)^2 - (2/2)^2$ = **0.0**

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Debt	Income	Age	Home owner
High	High	7	No
		(7 + 12) / 2 = 9.5	
High	Low	12	No
		15	
Low	High	18	Yes
		26.5	
Low	High	35	Yes
		36.5	
High	High	38	Yes
		44	
High	Low	50	No
		66.5	
Low	Low	83	No

Step 3: Calculate the **Gini impurity** for **feature Age.**

An example below shows how to calculate **Gini impurity** for **threshold Age < 44**



Homeowners	3
Not homeowners	2
Gini impurity	$= 1 - (3/5)^2 - (2/5)^2$ = **0.48**

Homeowners	0
Not homeowners	2
Gini impurity	$= 1 - (0/2)^2 - (2/2)^2$ = **0.0**

Weighted Gini impurity for feature **Age < 44**
$= (5/7) \times 0.48 + (2/7) \times 0.0 =$ **0.343**

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 3: Calculate the **Gini impurity** for **feature Age.**
We repeat the same calculations **for all Age averages**

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

(7 + 12) / 2 = 9.5 → **Weighted Gini impurity: 0.429**

15 → **Weighted Gini impurity: 0.343**

26.5 → **Weighted Gini impurity: 0.476**

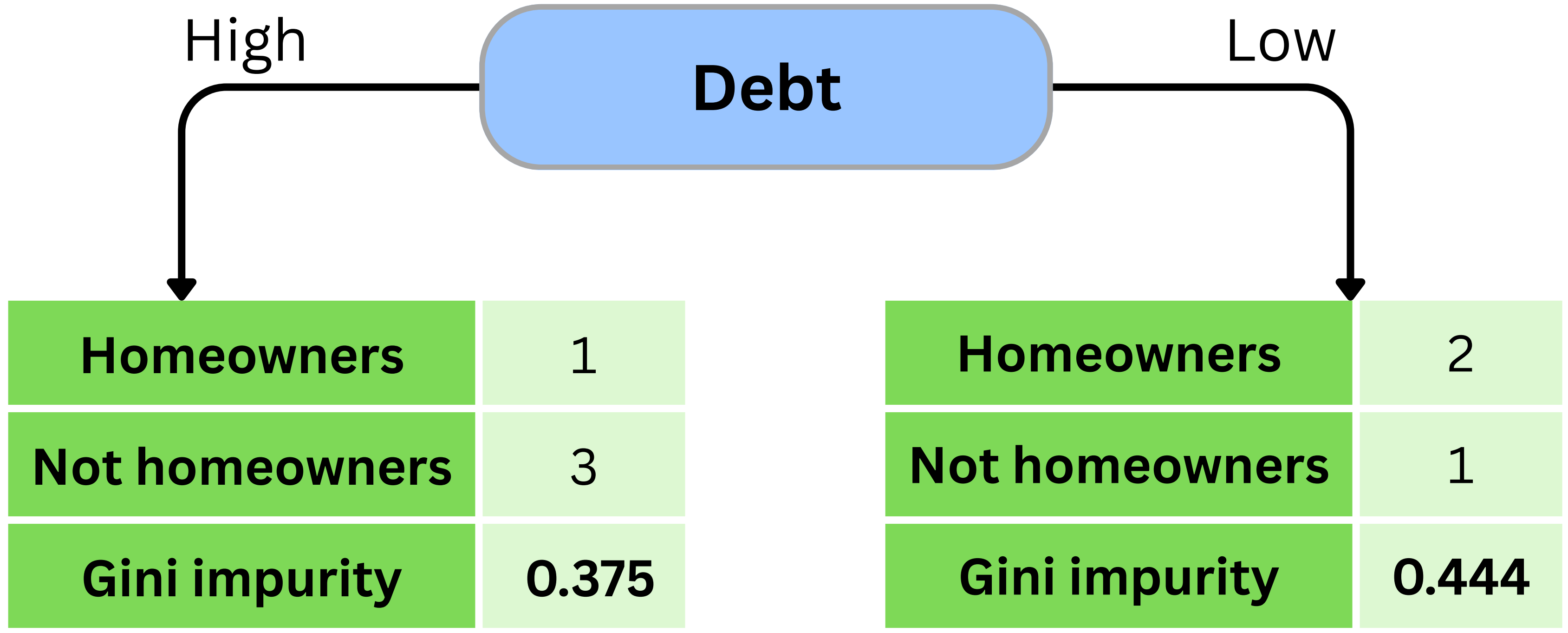
36.5 → **Weighted Gini impurity: 0.476**

44 → **Weighted Gini impurity: 0.343**

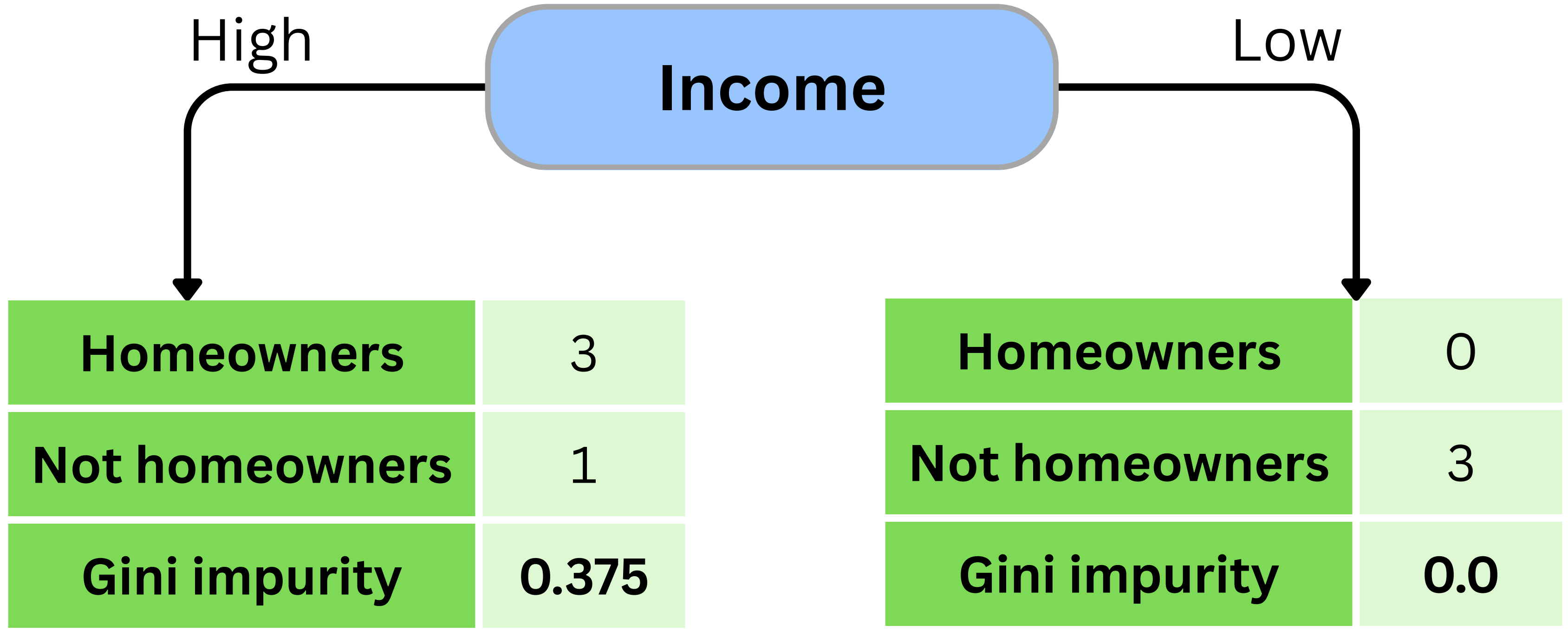
66.5 → **Weighted Gini impurity: 0.429**

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

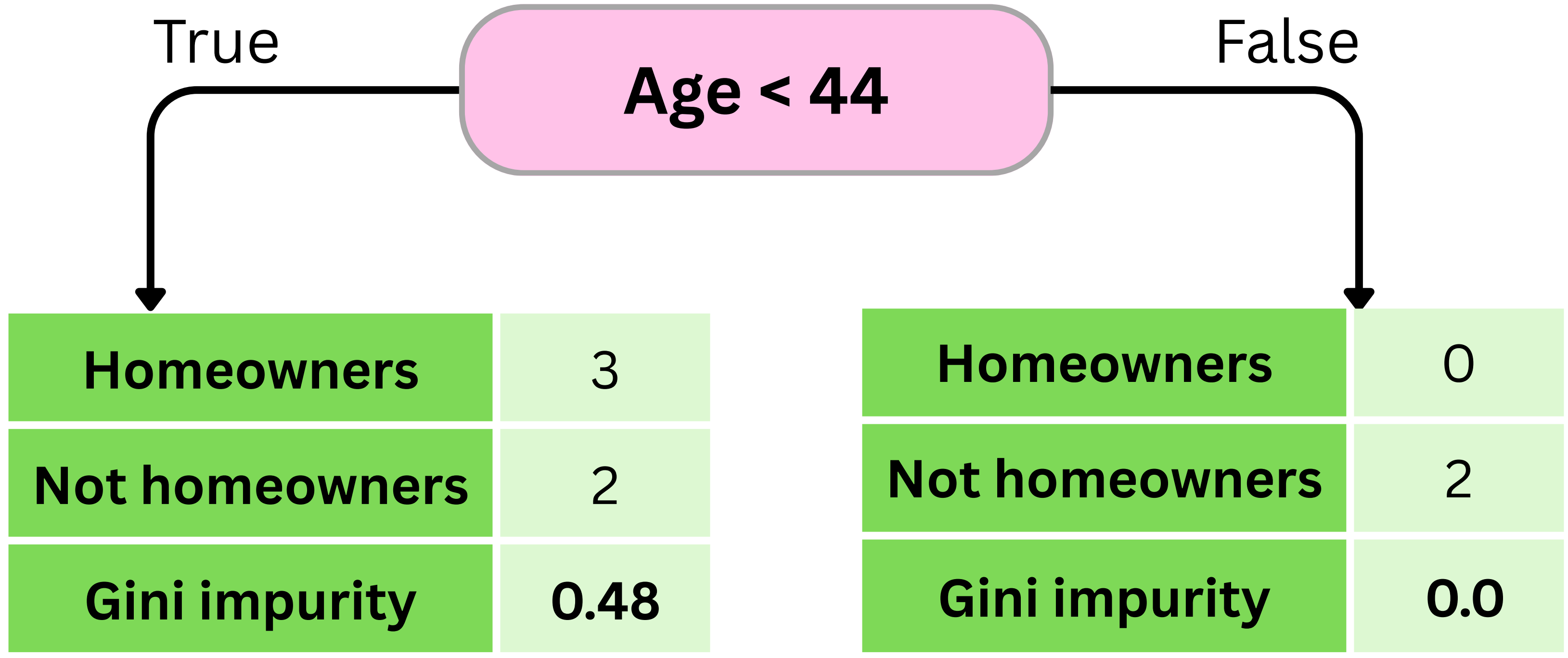
Step 4: Pick the feature with the lowest **Weighted Gini impurity**



Weighted Gini impurity for feature **Debt**
= **0.405**



Weighted Gini impurity for feature **Income**
= **0.214**

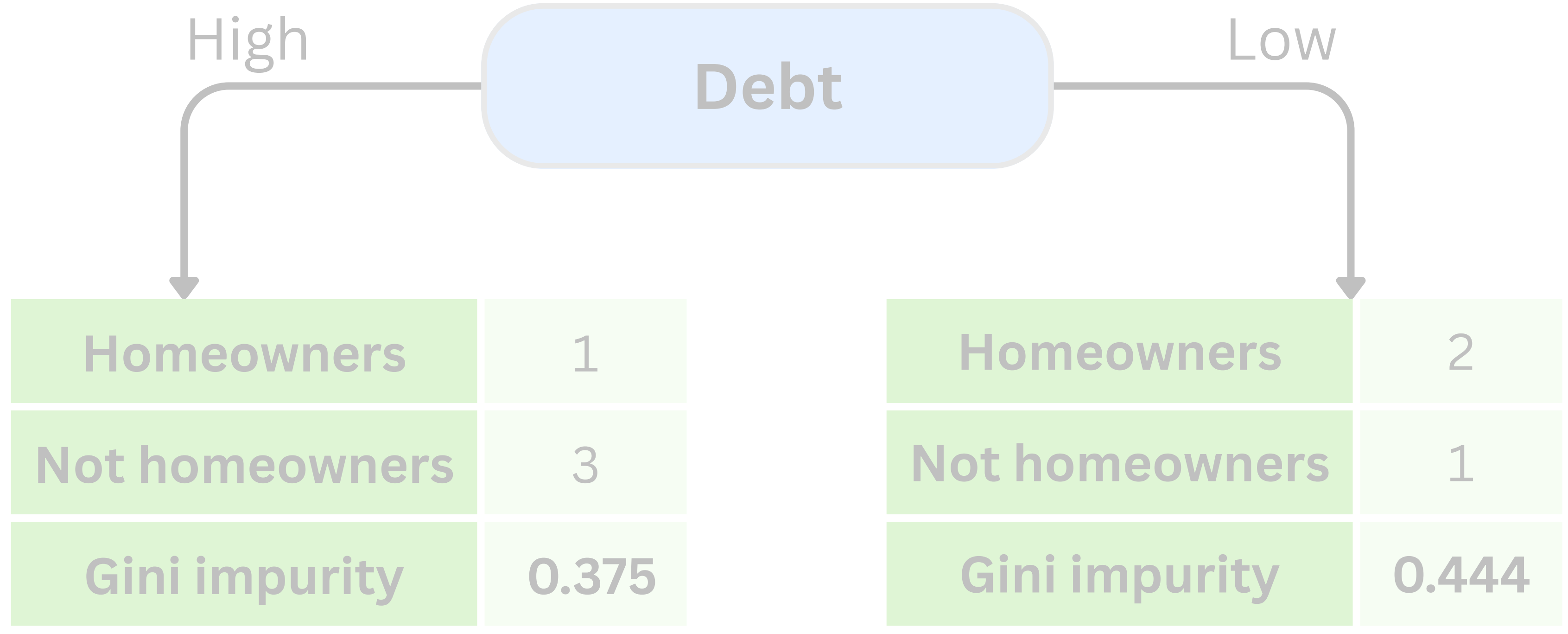


Weighted Gini impurity for feature **Age < 44**
= (5/7) × 0.48 + (2/7) × 0.0 = **0.343**

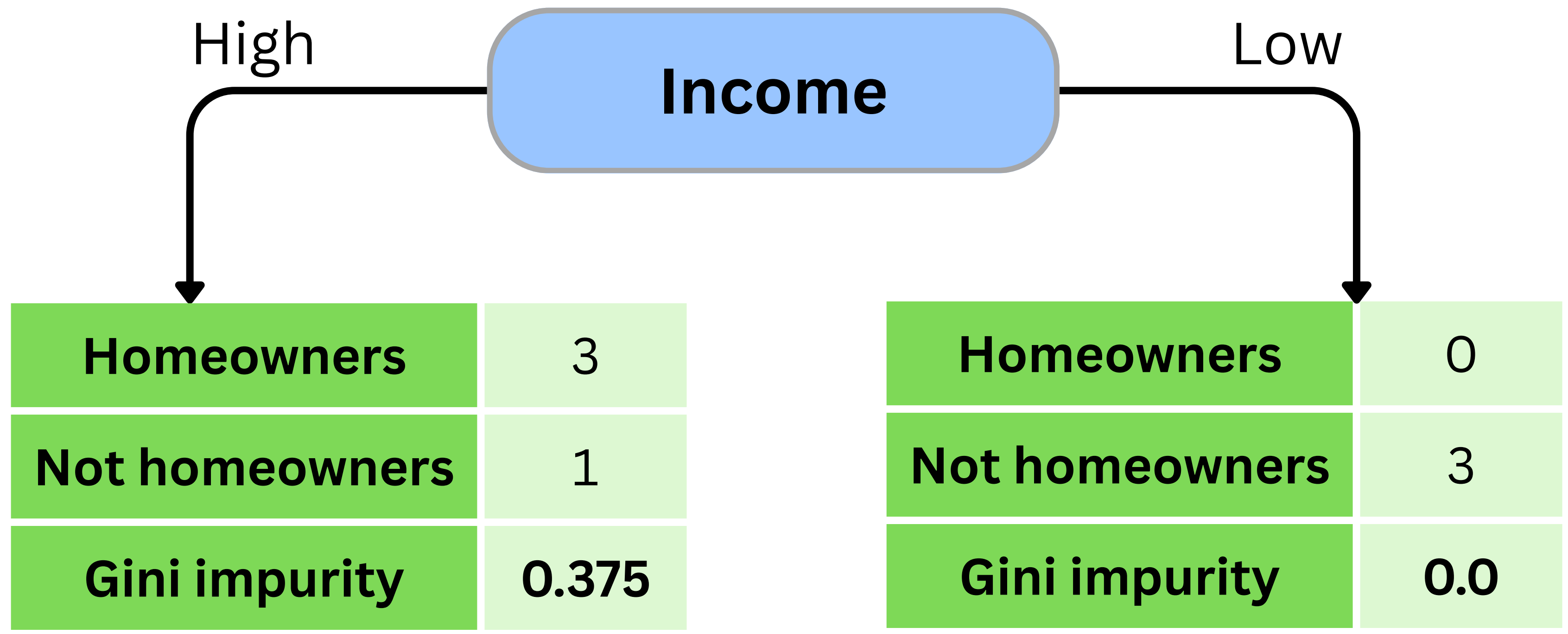
How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 4: Pick the feature with the lowest **Weighted Gini impurity**

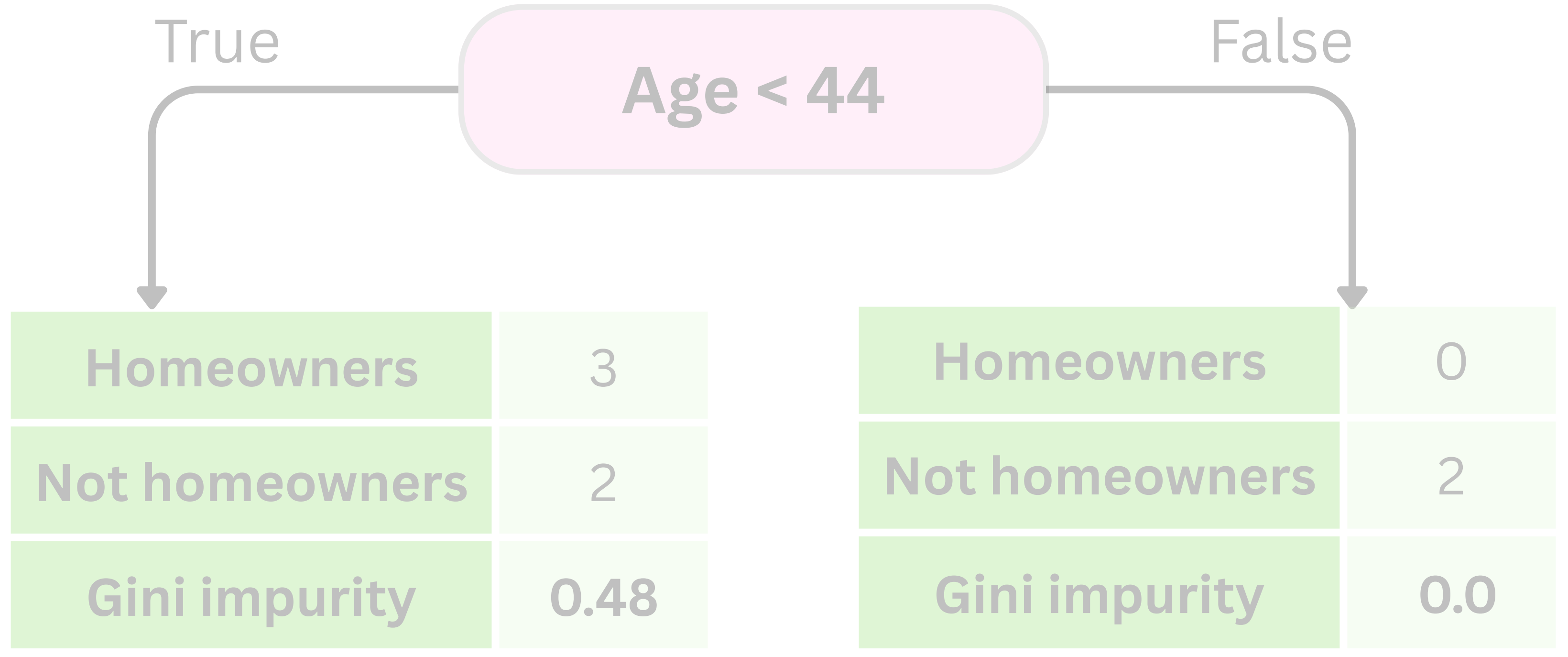
Since **Income** has the lowest Gini impurity, we pick it to be our root node



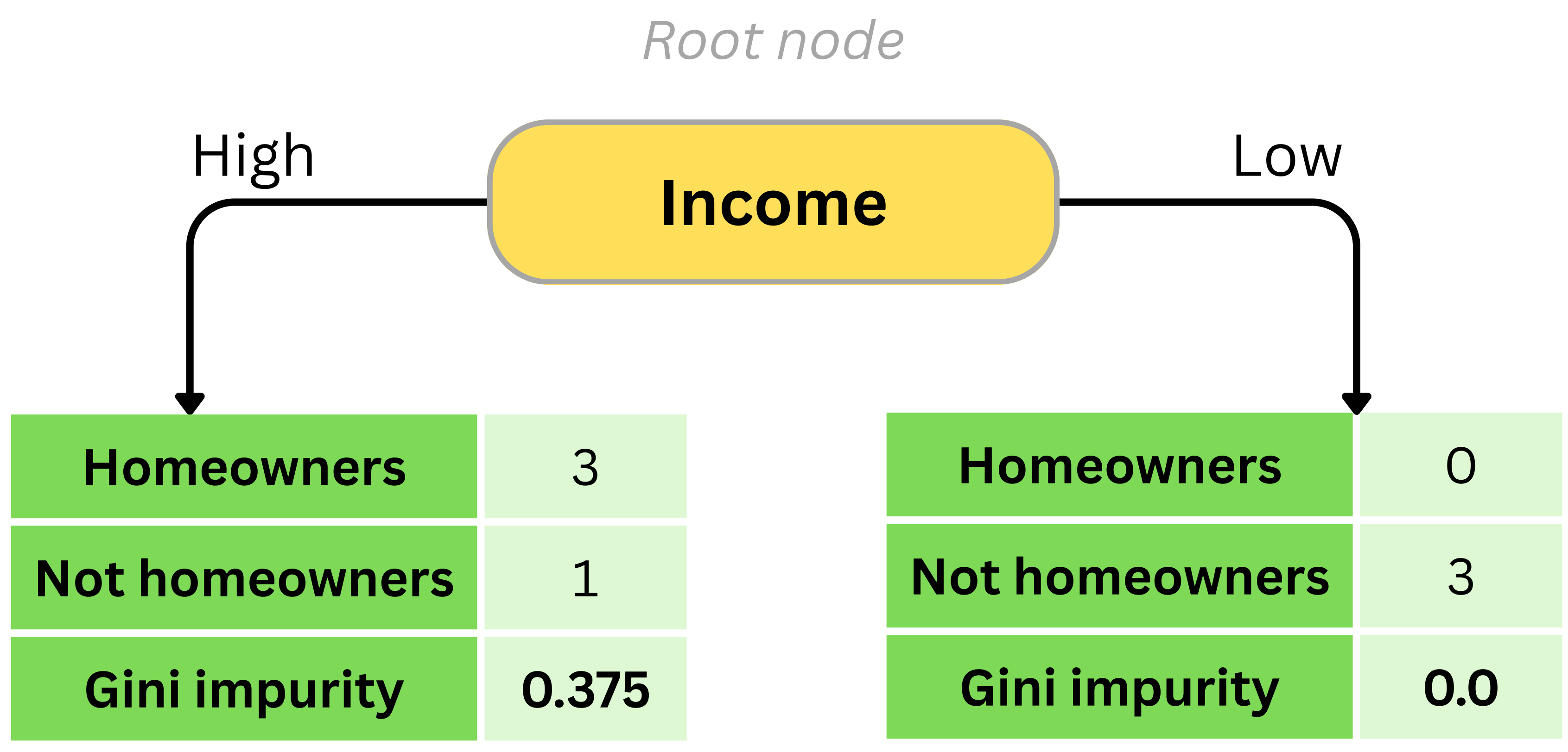
Weighted Gini impurity for feature **Debt** = **0.405**



Weighted Gini impurity for feature **Income** = **0.214**



Weighted Gini impurity for feature **Age < 44** = (5/7) × 0.48 + (2/7) × 0.0 = **0.343**



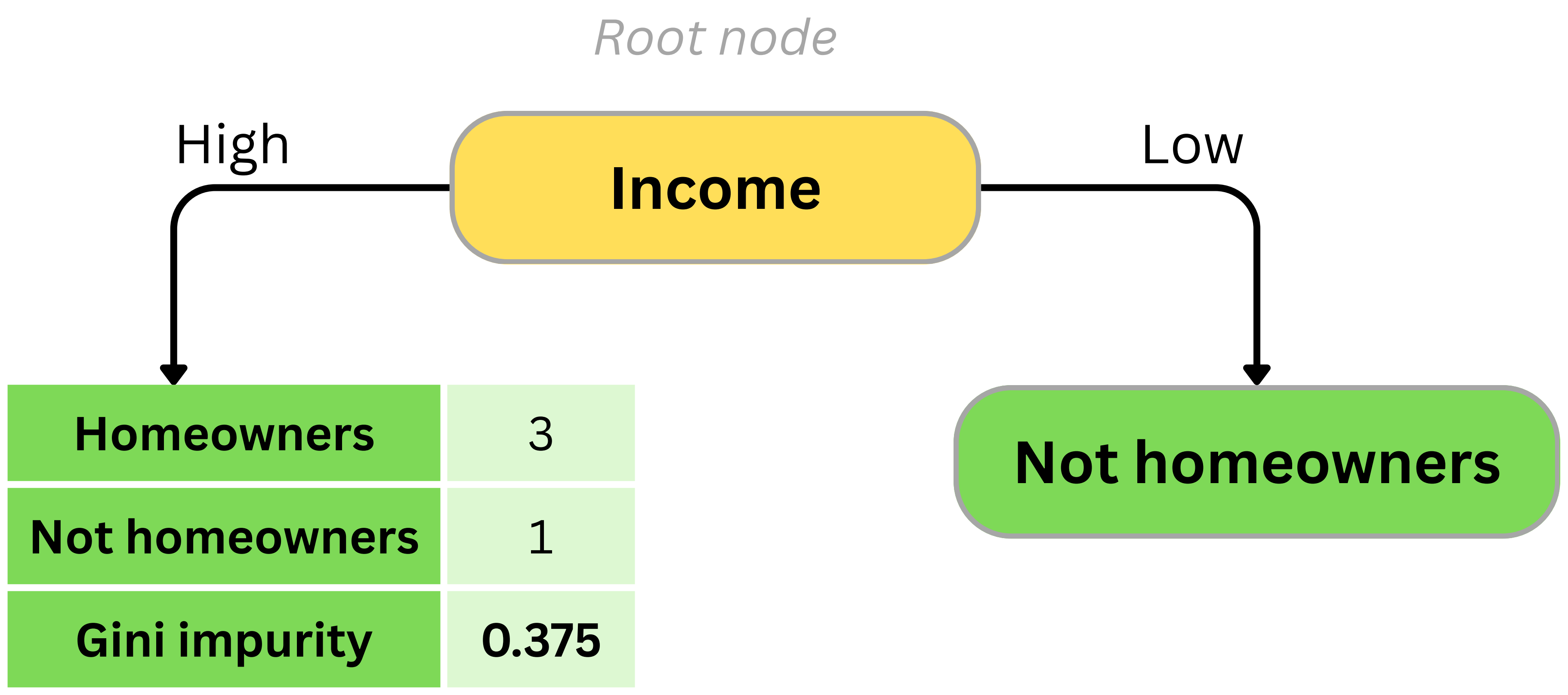
Weighted Gini impurity for feature **Income**
= **0.214**

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 4: Pick the feature with the lowest **Weighted Gini impurity**

Income has the lowest Weighted Gini impurity, so we pick it to be our root node

Bonus: Since the right child of **Income** is already completely pure, we can safely treat it as a leaf node and continue processing only the left child.



How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 4: Pick the feature with the lowest **Weighted Gini impurity**

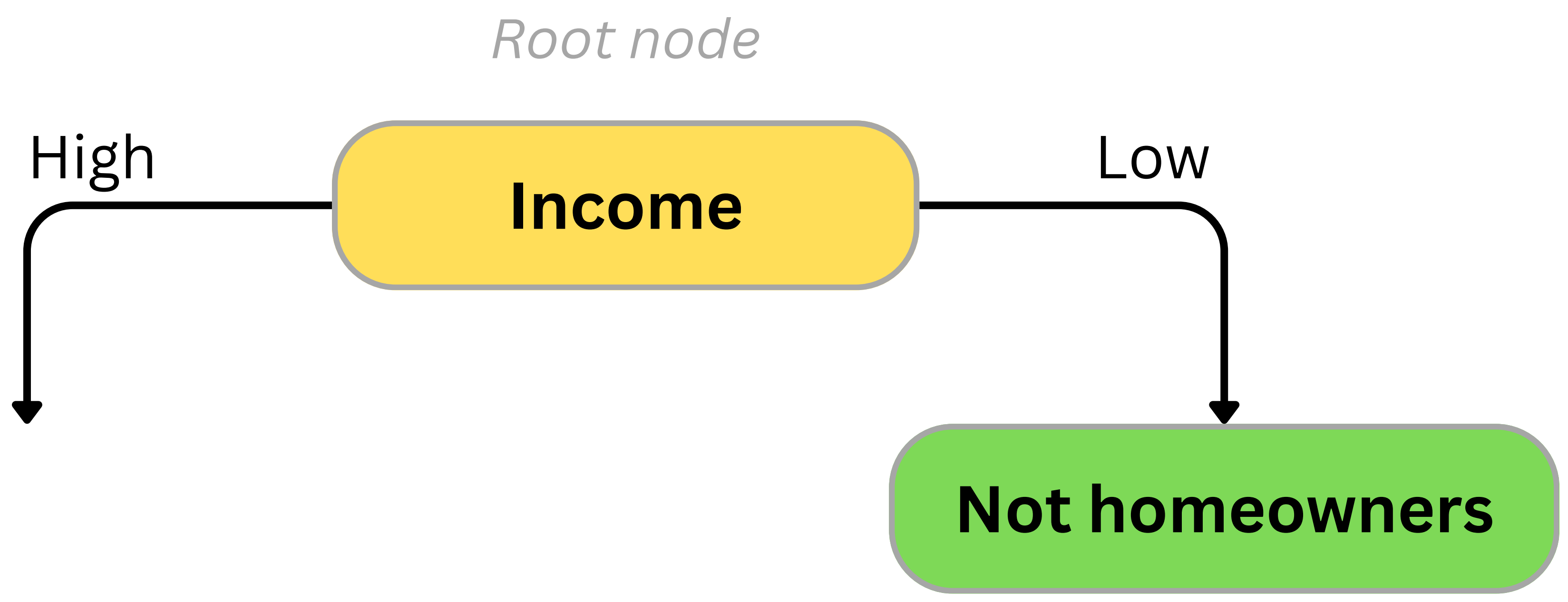
Income has the lowest Weighted Gini impurity, so we pick it to be our root node

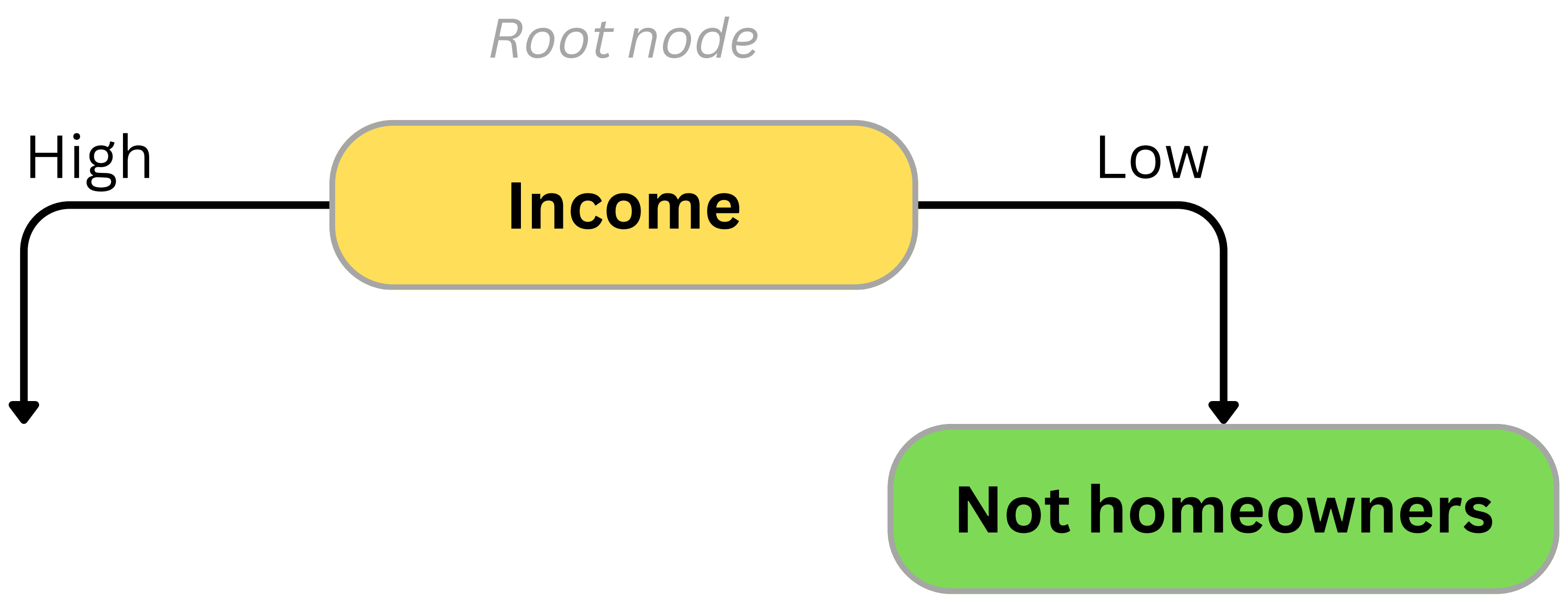
Bonus: Since the right child of **Income** is already completely pure, we can safely treat it as a leaf node and continue processing only the left child.

Debt	Income	Age	Home owner
High	High	7	No
High	Low	12	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes
High	Low	50	No
Low	Low	83	No

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 5: We repeat the process to find the optimal split for Income's left child, this time considering only data from individuals with high income.





Debt	Income	Age	Home owner
High	High	7	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

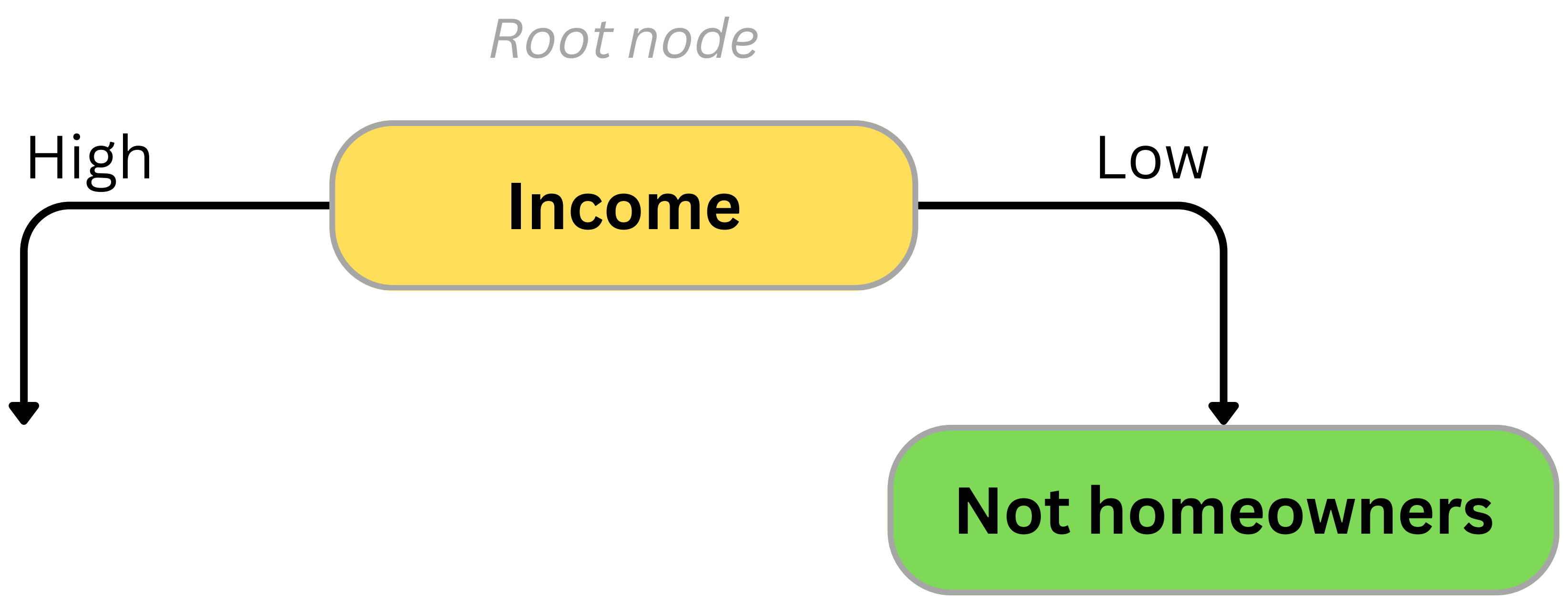
Step 5: We repeat the process to find the optimal split for Income’s left child, this time considering only data from individuals with high income.

We need to either **feature Debt** or **an Age threshold** to be our next node, again by calculating their Weighted Gini impurity.

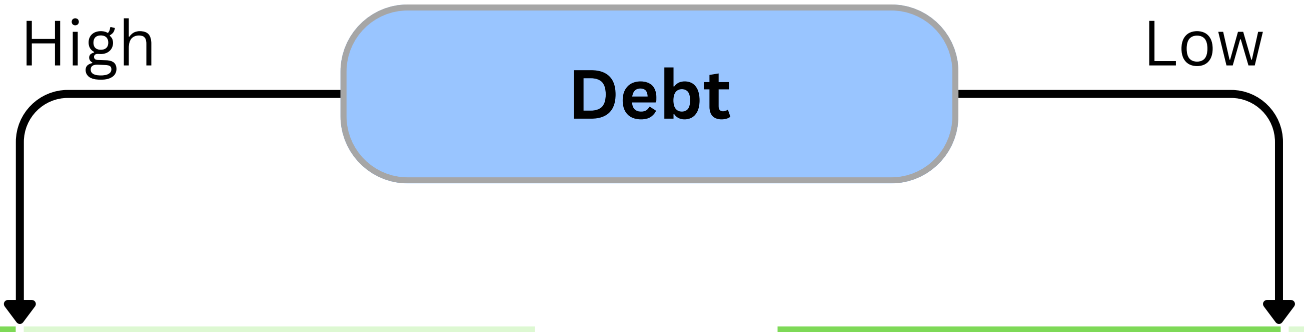
How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 5: We repeat the process to find the optimal split for Income's left child, this time considering only data from individuals with high income.

We need to either **feature Debt** or **an Age threshold** to be our next node, again by calculating their Weighted Gini impurity.



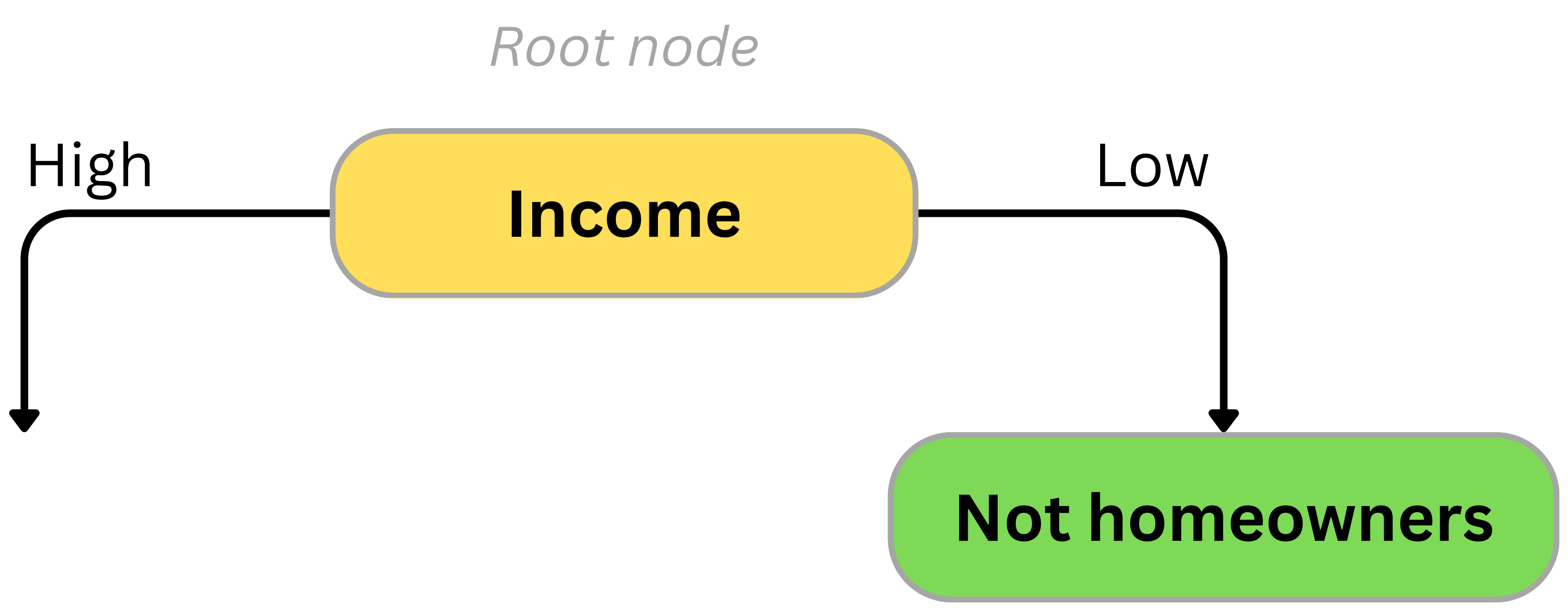
Debt	Income	Age	Home owner
High	High	7	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes



Homeowners	1
Not homeowners	1
Gini impurity	$= 1 - (1/2)^2 - (1/2)^2$ $=$ **0.5**

Homeowners	2
Not homeowners	0
Gini impurity	$= 1 - (2/2)^2 - (0/2)^2$ $=$ **0.0**

Weighted Gini impurity for feature **Debt**
$= (2/4) \times 0.5 + (2/4) \times 0.0 =$ **0.25**



How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 5: We repeat the process to find the optimal split for Income's left child, this time considering only data from individuals with high income.

We need to either **feature Debt** or **an Age threshold** to be our next node, again by calculating their Weighted Gini impurity.

Since **Age threshold < 12.5** has the lowest Gini impurity, we choose it to be our next branch

Debt	Income	Age	Home owner
High	High	7	No
Low	High	18	Yes
Low	High	35	Yes
High	High	38	Yes

12.5 → **Weighted Gini impurity: 0**

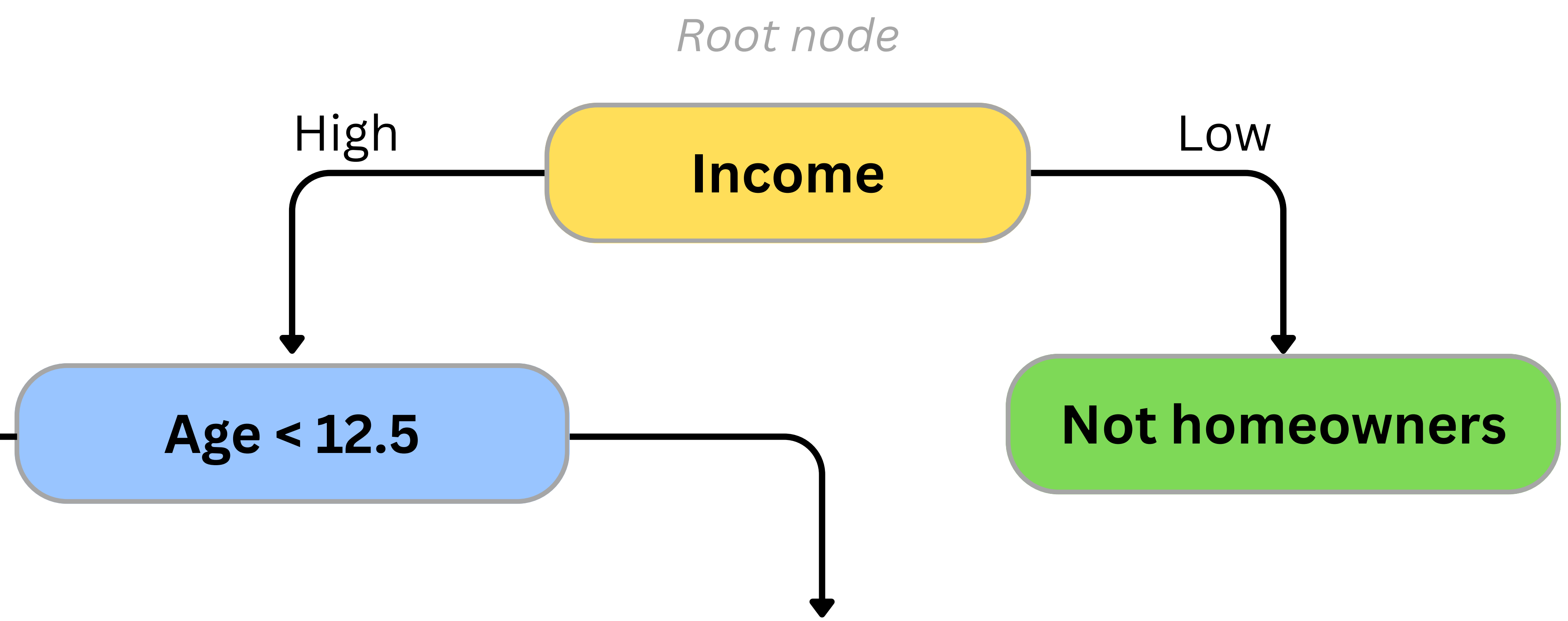
26.5 → **Weighted Gini impurity: 0.25**

36.5 → **Weighted Gini impurity: 0.333**

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 5: Since **Age threshold < 12.5** has the lowest Gini impurity, we choose it to be our next branch

Bonus: Since both children of **Age < 12.5** are completely pure, we can treat them as a leaf nodes and thus **finish our Decision Tree**.



Homeowners	0
Not homeowners	3
Gini impurity	0.0

Homeowners	3
Not homeowners	0
Gini impurity	0.0

How to build a Decision Tree that predicts whether someone is a **Homeowner**?

Step 5: Since **Age threshold < 12.5** has the lowest Gini impurity, we choose it to be our next branch

Bonus: Since both children of **Age < 12.5** are completely pure, we can treat them as a leaf nodes and thus **finish our Decision Tree**.